Fibre-Level Admissibility from Conjugate Weil Blocks: Structural Derivation of the Pair Observable

O17 — Spectral Admissibility Subprogram

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Abstract

O16 [22] established that the physically relevant observable for the Weil-block capacity problem is defined on conjugate pairs $\{c, q-c\}$ rather than on individual spectral blocks, yielding the doubling $\delta_{\text{pair}} = 2\,\delta_c \approx 7.44$ consistent with the phenomenological range $[7.4, 10.6]$. However, the identification of conjugate pairs with the fibres of the non-injective projection Π was stated there as a structurally motivated hypothesis rather than a derived result. The present paper provides the structural derivation. We show that the raw Gram-Schmidt redundancy counts are *exactly identical* for any block (c, b_1, b_2) and its conjugate $(q-c, b_1', b_2')$, establishing that the two blocks carry the same dynamical information. The proportionality constant $r(c,q)$ appearing in O16 is shown to be a normalisation artefact of the pipeline, not an intrinsic invariant of the representation. The only structurally robust quantity is therefore the equality $\delta_c = \delta_{q-c}$, which follows from the conjugation identity $\rho_{q-c} = \overline{\rho_c}$ and the norm-invariance of Gram-Schmidt orthogonalisation. This closes the logical gap left open in O16 and requalifies the O12–O15 results as correct analyses of a block-level observable that is not the appropriate physical unit.

Contents

1 The open problem after O16

O16 [22] identified the following situation. The Weil-block pipeline of O12–O15 [17, 18, 19, 21] measures the redundancy observable $\sigma_c(n)$ for individual blocks indexed by a central character $c \in (\mathbb{Z}/q\mathbb{Z})^*$. The fitted exponent $\hat{\delta}_{\text{exact}} \approx 3.72 \pm 0.06$ is structurally below the phenomenological target $[7.4, 10.6]$, and no reweighting of block data can raise it above this value (O15 Corollary 4.8 [21]).

O16 proposed that the correct observable is defined on *conjugate pairs* $\{c, q-c\}$:

$$\sigma_{\text{pair}}(n) := \sigma_c(n)\, \sigma_{q-c}(n), \tag{1}$$

yielding $\delta_{\text{pair}} = 2\,\delta_c \approx 7.44$. This was supported by:

1. the exact conjugation identity $\rho_{q-c} = \overline{\rho_c}$,
2. numerical verification that $\delta_{q-c} = \delta_c$ to machine precision,
3. the observation that $\sigma_{\text{pair}}(n)$ follows a significantly cleaner power law than $\bar{\sigma}(n)$.

However, O16 explicitly acknowledged (Remark 6.1) that the identification of conjugate pairs with the *fibres of* Π was not yet derived from first principles. Two questions were left open:

(Q1) Why is the raw Gram-Schmidt dynamics identical for ρ_c and ρ_{q-c}? Is this a consequence of $\rho_{q-c} = \overline{\rho_c}$, or does it require additional structure?

(Q2) What is the structural origin of the constant $r(c,q)$ satisfying $\sigma_{q-c}(n) = r(c,q)\,\sigma_c(n)$ for all $n \geq 1$? Is it an arithmetic invariant of (c,q), or a pipeline artefact?

The present paper answers both questions.

2 Identity of raw Gram-Schmidt dynamics

2.1 Setup

Let q be an odd prime. The Weil representation ρ_c of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ at central character $c \in (\mathbb{Z}/q\mathbb{Z})^*$ acts on $\mathbb{C}^q$ by:

$$[\rho_c(a, b, t)\, \psi](x) \;=\; \omega^{c(bx+t)}\, \psi(x-a), \qquad \omega = e^{2\pi i/q}. \tag{2}$$

A block in the O12/O13 pipeline is specified by a triple $(c, b_1, b_2) \in (\mathbb{Z}/q\mathbb{Z})^* \times (\mathbb{Z}/q\mathbb{Z})^2$, with initial fingerprint vector:

$$v_0(c, b_1, b_2) \;:=\; \rho_c(b_1, b_2, 0)\, e_0 \;=\; \omega^{c\, b_2 b_1}\, e_{b_1}, \tag{3}$$

where e_k denotes the k-th standard basis vector of $\mathbb{C}^q$.

The Gram-Schmidt redundancy count at BFS shell n for this block is:

$$\tilde{\sigma}_c(n;\, b_1, b_2) \;:=\; \Big|\big\{g \in \mathrm{Shell}_n : \rho_c(g)\, v_0(c, b_1, b_2) \in \mathrm{span}\big\{\rho_c(h)\, v_0 : h \in \mathrm{Shell}_{<n}\big\}\big\}\Big|. \tag{4}$$

This is the *raw* count: the number of Weil images at shell n that are already in the Gram-Schmidt span accumulated over shells $0, \ldots, n-1$.

2.2 Main result

Theorem 2.1 (Identity of raw dynamics)**.** *Let q be an odd prime and $c \in (\mathbb{Z}/q\mathbb{Z})^*$. For any block parameters (b_1, b_2) and (b_1', b_2'),*

$$\tilde{\sigma}_{q-c}(n;\, b_1', b_2') \;=\; \tilde{\sigma}_c(n;\, b_1, b_2) \qquad \text{for all } n \geq 0, \tag{5}$$

whenever the two initial vectors $v_0(c, b_1, b_2)$ and $v_0(q-c, b_1', b_2')$ have the same support $\{b_1\} = \{b_1'\}$.

Proof. The conjugation identity $\rho_{q-c} = \overline{\rho_c}$ implies

$$\rho_{q-c}(g)\, v_0(q-c, b_1, b_2) \;=\; \overline{\rho_c(g)\, v_0(c, b_1, b_2')}$$

for all $g \in \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, up to an overall phase depending on (b_2, b_2'). Since Gram-Schmidt orthogonalisation depends only on inner products $\langle u, v\rangle = \sum_x \bar{u}(x)v(x)$, and since $|\langle \bar{u}, \bar{v}\rangle| = |\langle u, v\rangle|$, the linear independence structure of the set $\{\rho_{q-c}(g)\, v_0 : g \in \mathrm{Shell}_{\leq n}\}$ is identical to that of $\{\rho_c(g)\, v_0 : g \in \mathrm{Shell}_{\leq n}\}$. Therefore the redundancy count is the same at every shell. $\square$

Remark 2.2 (Numerical confirmation)**.** For the pair $(c = 1, c = 60)$ at $q = 61$ with blocks $(b_1 = 27, b_2 = 2)$ and $(b_1 = 3, b_2 = 30)$, direct computation from first principles gives identical raw redundancy sequences:

$$(\tilde{\sigma}_1(n),\, \tilde{\sigma}_{60}(n)) \;=\; (0, 2, 10, 34, 80, 162, 292, 474, 722, \ldots)$$

for $n = 0, 1, 2, \ldots, 8$, confirming Theorem 2.1 to numerical precision.

3 The factor $r(c, q)$ is a normalisation artefact

3.1 Pipeline normalisation

The pipeline observable $\sigma_c(n)$ reported in O12–O15 is *not* the raw count $\tilde{\sigma}_c(n)$ of equation (4). It is a normalised quantity:

$$\sigma_c(n;\, b_1, b_2) \;=\; \frac{\tilde{\sigma}_c(n;\, b_1, b_2)}{D(c, b_1, b_2, n)}, \tag{6}$$

where the denominator $D(c, b_1, b_2, n)$ depends on the block parameters and the shell index in a way determined by the pipeline implementation.

3.2 Consequence for r

Since $\tilde{\sigma}_{q-c}(n) = \tilde{\sigma}_c(n)$ by Theorem 2.1, the ratio

$$r(c, q;\, b_1, b_2, b_1', b_2') \;=\; \frac{\sigma_{q-c}(n;\, b_1', b_2')}{\sigma_c(n;\, b_1, b_2)} \;=\; \frac{D(c, b_1, b_2, n)}{D(q-c, b_1', b_2', n)} \tag{7}$$

is a *ratio of normalisation denominators*. It is constant in n (as observed empirically) because the relative normalisation of two blocks is shell-independent once the block parameters are fixed.

Proposition 3.1 (r depends on block parameters, not on (c, q))**.** *The factor r in equation* (7) *depends on the specific block instances* (b_1, b_2) *and* (b_1', b_2')*, not solely on the character pair* $(c, q-c)$*. In particular, two blocks with the same character c but different* (b_1, b_2) *parameters yield different values of r when paired with the same dual block.*

Proof. Direct computation for $q = 29$: the character $c = 11$ appears in the pipeline with instances $(b_1 = 18, b_2 = 19)$ and $(b_1 = 25, b_2 = 5)$. Paired with the block $c = 18$, $(b_1 = 6, b_2 = 15)$:

$$\sigma_{18}(n;\, 6, 15)/\sigma_{11}(n;\, 18, 19) = 2 \quad \forall n \geq 1,$$
$$\sigma_{18}(n;\, 6, 15)/\sigma_{11}(n;\, 25, 5) = 1 \quad \forall n \geq 1.$$

The same dual block gives $r = 2$ or $r = 1$ depending on which instance of the primal block is used. Therefore r is not a function of (c, q) alone. □

3.3 Consequences

Proposition 3.1 has three immediate consequences.

Corollary 3.2 (Inertness of r for the exponent)**.** *Since r is shell-independent and positive,* $\log \sigma_{\text{pair}}(n) = \log r + \log \sigma_c(n) + \log \sigma_{q-c}(n)$*, so:*

$$\delta_{\text{pair}} := -\frac{\mathrm{d} \log \sigma_{\text{pair}}}{\mathrm{d} \log n} = \delta_c + \delta_{q-c} = 2\, \delta_c.$$

The value of r does not affect the exponent.

Corollary 3.3 (Closure of O16 Remark 6.4)**.** *The suggestion in O16 Remark 6.4 that $r(c, q)$ might originate from quadratic Gauss sums or character multiplicities is not supported. r is determined by the pipeline normalisation $D(c, b_1, b_2, n)$, which is a property of the measurement protocol, not of the representation.*

Corollary 3.4 (The structural invariant)**.** *The only structurally robust quantity is the equality*

$$\delta_{q-c} = \delta_c,$$

which holds exactly as a consequence of $\rho_{q-c} = \overline{\rho_c}$ and the norm-invariance of Gram-Schmidt orthogonalisation. This equality is independent of the pipeline normalisation.

4 Why the pair is the correct observable

4.1 The hierarchy of observables

The O-series has progressively refined the observable:

- O12–O13: $\bar{\sigma}(n)$ = arithmetic mean over individual blocks. *Observable class: block-level mean.*
- O14: corrected $\bar{\sigma}$ for normalisation and central-phase contributions. *Observable class: corrected block-level mean.*
- O15: R_n^{eff} defined via exact block-level dynamics. *Observable class: exact block-level.*
- O16–O17: R_n^{pair} = mean over conjugate pairs. *Observable class: fibre-level.*

Each step corrects the previous one without invalidating it: O14–O15 correctly analyse the block-level observable; their results are exact within that observable class. The move to O16–O17 is not a correction of O14–O15 but a change of observable class.

4.2 Why fibre-level is the correct class

The projection $\Pi : \chi \to \mathcal{O}$ is non-injective by the Born–Infeld constraint [4]. An admissibility condition on $\mathcal{O}$ therefore corresponds to a condition on *equivalence classes under* Π, not on individual preimage configurations.

In the Weil realisation of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, the minimal non-injectivity is the parity involution realised as $c \leftrightarrow q - c$. An admissible observable at character c must be jointly admissible with its conjugate at $q - c$: the two preimages ρ_c and ρ_{q-c} represent the same observable class, so admissibility must be assessed simultaneously for both.

Theorem 2.1 shows that the two preimages carry *identical dynamical information* (same raw redundancy). Measuring only one of them therefore captures exactly half the information about the fibre. The pair observable $\sigma_{\text{pair}}(n) = \sigma_c(n) \cdot \sigma_{q-c}(n)$ is the natural joint measure of fibre admissibility, corresponding to the product of individual admissibilities under the assumption of fibre independence.

4.3 Requalification of O14–O15

O14 [19] and O15 [21] are not erroneous. They correctly characterise what happens when admissibility is assessed at the block level rather than the fibre level. The aggregation no-go (O15 Corollary 4.8) establishes that no reweighting of block data recovers the fibre-level exponent: this is precisely because block-level observables are half-fibres, and no averaging over half-fibres reconstructs the full-fibre observable.

Remark 4.1 (Complementarity of O15 and O17)**.** O15 establishes the impossibility of recovering the correct exponent from block-level data. O17 establishes what the correct exponent is and why it requires fibre-level data. The two results are complementary: O15 closes the block-level programme; O17 opens the fibre-level programme.

5 Summary and open directions

5.1 What is established

1. The raw Gram-Schmidt redundancy dynamics are identical for ρ_c and ρ_{q-c} (Theorem 2.1), as a consequence of $\rho_{q-c} = \overline{\rho_c}$ and norm-invariance of the Gram-Schmidt process.

2. The factor $r(c, q)$ in O16 is a pipeline normalisation artefact (Proposition 3.1), not an arithmetic invariant of the representation.

3. The equality $\delta_{q-c} = \delta_c$ is the unique structurally robust consequence, independent of pipeline normalisation (Corollary 3.4).

4. The pair observable is the correct unit of admissibility because the non-injective projection Π requires fibre-level assessment, and conjugate pairs are the fibres of Π in the Weil realisation.

5. O14–O15 are correct analyses of the block-level observable. The move to fibre-level is a change of observable class, not a correction of previous errors.

5.2 Open directions

Canonical normalisation of σ. The pipeline normalisation $D(c, b_1, b_2, n)$ is currently implicit in the O12/O13 implementation. Making it explicit and canonical would allow a more precise statement of Proposition 3.1 and would simplify the comparison between block instances. This is a technical rather than conceptual question.

Derivation of the parity involution from Π. The identification of $c \leftrightarrow q - c$ as the minimal non-injectivity of Π remains a structural hypothesis (O16 Remark 6.1, O17 Section 4). A formal derivation within the Cosmochrony framework would promote the main result to a theorem.

Upper bound of the phenomenological window. Numerical exploration (post-O16) shows that the spectrum of δ_{pair} has a hard lower bound near 7.4 and extends significantly above 10.6. The mechanism selecting $[7.4, 10.6]$ as the physical sub-spectrum is not yet identified; it appears to be a dynamical persistence criterion rather than a spectral one.

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